
AirSense

Campus Air Quality Intelligence Pilot

A joint initiative of BIC Islamabad and BIC Karachi, developed through the BIC x University of Essex COIL AI Hackathon

SUBMITTED TO

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SUBMITTED BY

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Executive Summary

AirSense is an AI-powered air quality prediction and decision support platform, developed jointly by student teams at BIC Islamabad and BIC Karachi through the BIC x University of Essex COIL AI Hackathon. What began as a hackathon submission has grown into a working system with a validated forecasting model and a clear path to real institutional and commercial use.

We are requesting approval for a three month campus pilot at BIC Islamabad, together with a one-time budget of Rs 20,000, to install a rooftop air quality sensor and collect the real-world data our models need. An identical pilot is running in parallel at BIC Karachi under the same leadership structure, so this is a joint, two-campus effort from day one, not a single-campus project.

For a modest, one-time investment, BIC gains a live applied research project, a public-facing air quality dashboard, and ownership of a system with genuine potential to grow into a commercial product carrying the BIC name.

Where This Began

AirSense began at the BIC x University of Essex COIL AI Hackathon, a Collaborative Online International Learning programme connecting BIC students with University of Essex students on a shared applied AI challenge. Our entry stood out for treating air quality as a decision problem rather than another monitoring dashboard. Using historical Islamabad weather data, we built and validated a forecasting model that explains 41 percent of PM2.5 variance with an average error of 49 micrograms per cubic metre, a strong result for a model trained entirely on downloaded, historical data. The project has since grown, under joint leadership at both campuses, Munim Qureshi at Islamabad and Areesha Aqeel at Karachi, into a full initiative with real institutional and commercial potential.

Why It Matters

Pakistan is ranked the most polluted country in the world by IQAir's 2025 World Air Quality Report, and the World Bank estimates air pollution costs the national economy close to 47 billion US dollars a year. Despite this, institutions and businesses routinely make decisions about outdoor activity, operations, and health without access to real-time, local air quality information. AirSense exists to close that gap, starting with the campuses that built it.

What We Are Proposing

We are requesting approval to install one IoT air quality sensor on the BIC Islamabad rooftop for three months, continuously logging PM2.5 and key weather variables. This is a real, physically deployed IoT system, not a simulation, and it lets us retrain our existing in-house models on genuine local data instead of historical downloads alone. BIC Karachi is running an identical deployment in parallel under Areesha Aqeel's leadership, using the same hardware and software, so both campuses build one shared system together.

Why Both Campuses Benefit

Benefit Area	What BIC Gains
Applied Research Asset	Three months of real, campus generated environmental data, usable across AI, Computer Science, and Environmental Science coursework at both campuses.

Cross Campus Collaboration	A single system co-built by Islamabad and Karachi together, demonstrating institute-wide capability rather than one campus acting alone.
Public Dashboard	A free, publicly accessible live air quality dashboard for the whole BIC community, and eventually the public, at no cost to either campus.
Institutional Positioning	BIC becomes one of the first Pakistani institutes with a deployed, student-built, AI-driven environmental monitoring system, at two campuses at once.
Zero Ongoing Cost	All software, modelling, and dashboard infrastructure is already built and maintained by the team. The one-time hardware cost is the full financial commitment.

Budget Request

We are requesting a one-time release of Rs 20,000 to procure and install pilot hardware for the BIC Islamabad deployment. This is a one-time cost with no software licensing fees, subscriptions, or recurring charges, since our full software stack is already built and maintained by the team.

Item	Cost (Rs)
Sensor unit and microcontroller	12,500
Weatherproof enclosure and rooftop mounting	2,500
Power supply and wiring	2,000
Data connectivity for the pilot period	2,000
Contingency (approximately 5 percent)	1,000
Total	20,000

Where This Goes Next

The campus pilot is Phase 1 of a three-phase roadmap that ends in a genuine commercial product.

Phase	Description	Key Output
Phase 1 Campus Pilot	Deploy sensors at both campuses, collect three months of data, retrain existing in-house models.	Validated two-city dataset and forecasting models.
Phase 2 Multi-Campus Refinement	Improve models using combined Islamabad and Karachi data; expand the decision framework.	Higher-accuracy models and a reusable blueprint for new cities.
Phase 3 Operational Platform	Launch as a subscription service turning forecasts into direct operational guidance for business and institutional clients.	Revenue-generating platform with BIC as its founding institution.

The commercial opportunity is significant. In Lahore alone, over 8,000 logistics companies are potential clients, and our projections show the platform breaking even at just 10 subscribers, with an average return of roughly 45 times subscription cost for a typical client through avoided losses. Every part of that opportunity traces back to data collected on BIC's own rooftops, which positions BIC as the founding institution behind this platform, regardless of how large it eventually grows.

Timeline

Stage	Activity
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Week 1	Budget release and vendor procurement
Week 2	Sensor installation and calibration at both campuses
Weeks 3-14	Continuous data collection, approximately three months
Final 2 weeks	Model retraining, validation, and completion report to management

Our Request

We respectfully request:

1. Approval of a one-time Rs 20,000 hardware budget for the BIC Islamabad sensor deployment
2. Permission to install the sensor at an agreed rooftop location for a three-month pilot period
3. Facilitation of rooftop access and connectivity support from campus facilities
4. Recognition and support for the parallel BIC Karachi deployment led by Areesha Aqeel

We are happy to present this proposal in person and answer any questions. Thank you for your continued support of student-led innovation at Beaconhouse International College.

Respectfully submitted,

Role	Name	Signature	Date
Head of Institute, BIC Islamabad	Ms. Saba Ahson		
Head of CSSE/AI, BIC Islamabad	Ms. Sahifa Alam		
Head of CSSE/AI, BIC Karachi	Mr. Sajid		
Project Lead, BIC Islamabad	Munim Qureshi		
Project Lead, BIC Karachi	Areesha Aqeel		